

Fig. 8.9 Limit of a function

When compared to the case of a function of single variable, for a function of two variables, there is a subtle depth in the limiting process. Here the values of $F(x,y)$ should approach the same value L, as (x,y) approaches (u,v) along **every possible path to** (u,v) (including paths that are not straight lines). Fig.8.9 explains the limiting process.

All the rules for limits (limit theorems) for functions of one variable also hold true for functions of several variables.

Now, following the idea of continuity for functions of one variable, we define continuity of a function of two variables.

Definition 8.7 (Continuity)

Suppose that $A=\{(x,y)|a<x<b, c<y<d\}\subset\mathbb{R}^2, F:A\to\mathbb{R}$. We say that F is continuous at (u,v) if the following hold :

(1) F is defined at (u,v) (2) $\lim\limits_{(x,y)\to(u,v)} F(x,y)=L$ exists (3) $L=F(u,v)$.

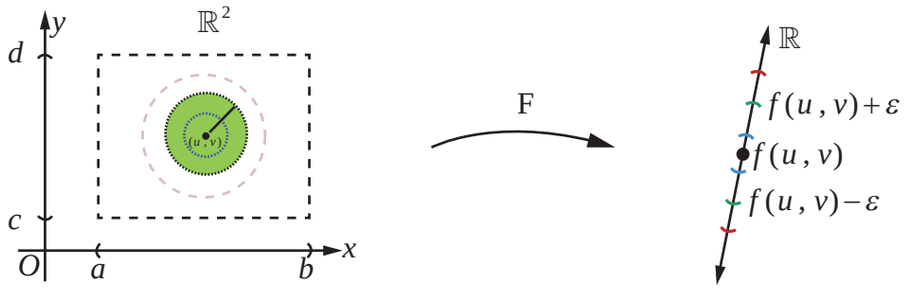


Fig. 8.10 Continuity of a function

Remark

(1) In Fig. 8.10 taking $L=F(u,v)$ will illustrate continuity at (u,v).

(2) Continuity for $f(x_1,x_2,\ldots,x_n)$ is also defined similarly as defined above.

Let us consider few examples as illustrations to understand continuity of functions of two variables.

Example 8.8

Let $f(x,y)=\dfrac{3x-5y+8}{x^2+y^2+1}$ for all $(x,y)\in\mathbb{R}^2$. Show that f is continuous on $\mathbb{R}^2$.

12th_Maths_Vol 2_EM_CH 8_Differentials and Partial.indd 71 16-12-2022 16:32:31